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Introduction to Optimization

Mathematical Programming

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About the Course

- Credit hours: 48
- Textbook: *Mathematical Programming: An Introduction to Optimization*, by Melvyn W. Jeter
- Reference: *Convex Optimization*, by Stephen Boyd
- Performance Assessment:
 - 1 Class attendance (10%)
 - 2 Assignments (40%)
 - 3 Final Exam (50%)



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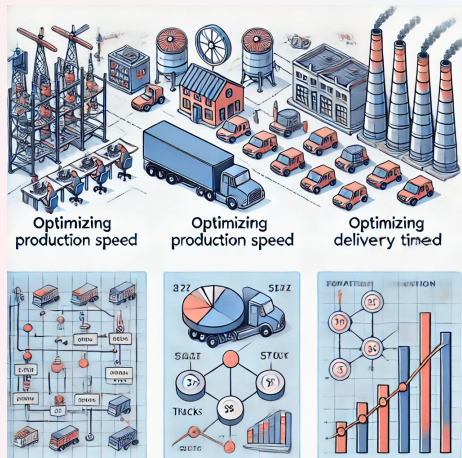


Introduction

- An introduction to optimization for students with background of linear algebra, calculus, and multivariable calculus
- Optimization is used everywhere nowadays with growing importance
- Applications:
 - Economic analysis
 - Management
 - Engineering technology
 - Military operations
 - etc



Examples



Examples



The Basic Mathematical Programming Problem

■ Definition:

$$\min f(x_1, x_2, \dots, x_n) \quad \text{or} \quad \max f(x_1, x_2, \dots, x_n) \quad (1)$$

$$\text{s.t.} \quad (x_1, x_2, \dots, x_n) \in \Omega \quad (2)$$

where

- $f(x_1, x_2, \dots, x_n)$ in (1) is the cost (objective) function.
- Ω in (2) is the set of feasible solutions.
- Unconstrained problem: $\Omega = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$.
- Constrained problem: Ω is a proper subset of $\{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$.



Linear Programming Problem

- Linear cost function:

$$f(x_1, \dots, x_n) = c_1x_1 + \dots + c_nx_n, \quad (3)$$

where $c_1, \dots, c_n$ are real constants. c_i is called the cost coefficient of the variable x_i .

- An equality constraint:

$$a_1x_1 + \dots + a_nx_n = b, \quad (4)$$

where $a_1, \dots, a_n$ are real constants.

- A linear inequality constraint:

$$a_1x_1 + \dots + a_nx_n \leq b, \quad (5)$$

- A problem that is NOT linear programming: nonlinear programming problem.
- Generally, linear programming problems are easier than nonlinear ones.



Example 1

- A manufacturer produces n products $P_1, P_2, \dots, P_n$ from m raw materials $M_1, \dots, M_m$. The amount of each raw material required for each product:

	P_1	P_2	$\dots$	P_n
M_1	$a_{1,1}$	$a_{1,2}$	$\dots$	$a_{1,n}$
M_2	$a_{2,1}$	$a_{2,2}$	$\dots$	$a_{2,n}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
M_m	$a_{m,1}$	$a_{m,2}$	$\dots$	$a_{m,n}$

- Available: b_i units of M_i , $\forall i = 1, \dots, m$.
- Revenue: c_j dollars for each unit of P_j sold.
- Question: Desire to maximize the total daily revenue.



Example 1

- x_j : # of product P_j that has been produced.
- $a_{i,j}x_j$: # of material M_i used to produce x_j units of P_j .
- # of the daily revenue:

$$c_1x_1 + \dots + c_nx_n \quad (6)$$

- # of material M_i consumed in total:

$$a_{i,1}x_1 + \dots + a_{i,n}x_n \leq b_i, \quad (7)$$

Example 1

- The cost function:

$$\max_{x_j, \forall j \in [1, n]} \{c_1 x_1 + \dots + c_n x_n\} \quad (8)$$

subject to

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n &\leq b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n &\leq b_2, \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n &\leq b_m, \\ x_1 \geq 0, x_2 \geq 0, \dots, x_n &\geq 0. \end{aligned} \quad (9)$$

Example 1

- Continuous variable: allowed to take on all fractional as well as integral values within its range of variation
- Discrete variable: must take on values from a discrete set of values
- Slack variable: inequality can be converted to an equality constraint by adding a non-negative variable
 - Take (9) for example:

$$a_{1,1}x_1 + \dots + a_{1,n}x_n + x_{n+1} = b_1,$$

x_{n+1} : Slack variable, assigned a cost coefficient of $c_{n+1} = 0$

Example 1

- The cost function (8) can be reformulated as

$$\max_{x_j, \forall j \in [1, n+m]} \underbrace{\{c_1 x_1 + \dots + c_n x_n + c_{n+1} x_{n+1} + \dots + c_{n+m} x_{n+m}\}}_{=\max_{x_j, \forall j \in [1, n]} \{c_1 x_1 + \dots + c_n x_n\}}$$

subject to

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n + x_{n+1} &= b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n + x_{n+2} &= b_2, \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n + x_{n+m} &= b_m, \\ x_1 \geq 0, \dots, x_n \geq 0, \quad x_{n+1} \geq 0, \dots, x_{n+m} &\geq 0. \end{aligned}$$



Example 2

- 4 salesmen + 4 sales districts
- The sales productivity ratings for each salesman S_i in each sales districts D_j :

	D_1	D_2	D_3	D_4
S_1	5	10	8	2
S_2	3	4	2	6
S_3	9	5	4	5
S_4	5	5	6	4

- Question: Determine the best assignment of salesman to sales districts.

Example 2

- Define

$$x_{i,j} = \begin{cases} 1, & \text{if } S_i \text{ assigned to } D_j, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

- The value of an assignment:

$$\begin{aligned} f(x_{i,j}) = & 5x_{1,1} + 10x_{1,2} + 8x_{1,3} + 2x_{1,4} + 3x_{2,1} + 4x_{2,2} + 2x_{2,3} + 6x_{2,4} \\ & + 9x_{3,1} + 5x_{3,2} + 4x_{3,3} + 5x_{3,4} + 5x_{4,1} + 5x_{4,2} + 6x_{4,3} + 4x_{4,4}. \end{aligned}$$

- Each salesman must be assigned to exactly one district:

$$\sum_{j=1}^4 x_{i,j} = 1, \quad \forall i. \quad (11)$$

- Each district is assigned exactly one salesman:

$$\sum_{i=1}^4 x_{i,j} = 1, \quad \forall j. \quad (12)$$

Example 2

■ Problem:

$$\max_{\forall i,j} \{f(x_{i,j})\}$$

subject to

$$\begin{aligned}x_{i,1} + x_{i,2} + x_{i,3} + x_{i,4} &= 1, & \forall i, \\x_{1,j} + x_{2,j} + x_{3,j} + x_{4,j} &= 1, & \forall j, \\0 \leq x_{i,j} &\leq 1, & \forall i, j.\end{aligned}$$

Example 3

- A flour company has 3 production lines which can produce A and B. The number of ten pound sacks of flour which can be produced daily by each of the production lines is summarized in the following table:

	L_1	L_2	L_3
A	15000	10000	12000
B	12000	13500	14500

- Question: Produce as many ten pound bags of flour of brands A and B as possible.



Example 3

- Define

$$x_i = \begin{cases} 1, & \text{if } L_i \text{ assigned to A,} \\ 0, & \text{otherwise,} \end{cases} \quad w_i = \begin{cases} 1, & \text{if } L_i \text{ assigned to B,} \\ 0, & \text{otherwise.} \end{cases}$$

- A production line cannot be assigned to both products:

$$x_1 w_1 = 0, \quad x_2 w_2 = 0, \quad x_3 w_3 = 0.$$

- All variables are non-negative



Example 3

- The above two constraints can be compressed into one:

$$\sum_{i=1}^3 x_i w_i = x_1 w_1 + x_2 w_2 + x_3 w_3 = 0. \quad (13)$$

- To insure that each production line is assigned to a job:

$$x_1 + w_1 = 1,$$

$$x_2 + w_2 = 1,$$

$$x_3 + w_3 = 1.$$

Example 3

- A nonlinear programming problem:

$$\max_{\forall x_i, w_j} \{15000x_1 + 10000x_2 + 12000x_3 + 12000w_1 + 13500w_2 + 14500w_3\}$$

subject to

$$x_1 + w_1 = 1,$$

$$x_2 + w_2 = 1,$$

$$x_3 + w_3 = 1,$$

$$\sum_{i=1}^3 x_i w_i = 0, \quad (\text{complementary constraint})$$

$$0 \leq x_i \leq 1, \quad 0 \leq w_i \leq 1, \quad \forall i.$$

Example 4

- Quadratic (nonlinear) programming problem:

$$\max_{\forall i,j} \left\{ c_1 x_1 + \dots + c_n x_n + \sum_{i=1}^n \sum_{j=1}^n b_{ij} x_i x_j \right\}$$

subject to

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1,$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2,$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m,$$

$$x_1 \geq 0, \quad x_2 \geq 0, \dots, \quad x_n \geq 0.$$

Example 4

- The portfolio selection problem:

$$\max_{\forall i,j} \left\{ c_1 x_1 + \dots + c_n x_n - \sum_{i=1}^n \sum_{j=1}^n b_{i,j} x_i x_j \right\}$$

subject to

$$\begin{aligned} x_1 + \dots + x_n &= 1, \\ x_1 \geq 0, \dots, x_n &\geq 0. \end{aligned}$$

- x_i : investment
- c_i : expected rate of return
- $\sum_{i=1}^n \sum_{j=1}^n b_{i,j} x_i x_j$: risk of investment



Global And Local Solutions

- Standard Euclidean distance:

$$\|x - \bar{x}\| = \sqrt{\sum_{i=1}^n (x_i - \bar{x}_i)^2}$$

- Neighborhood of $\bar{x}$:

$$N_{\varepsilon}(\bar{x}) = \{x : \|x - \bar{x}\| < \varepsilon\}$$

where ε is a positive real number.



Global And Local Solutions

- Local minimum (at $\bar{x} \in \Omega$):

$$\exists \varepsilon > 0, \text{ such that } f(\bar{x}) \leq f(x), \quad \forall x \in \Omega \cap N_\varepsilon(\bar{x}).$$

- Strict local minimum:

$$\exists \varepsilon > 0, \text{ such that } f(\bar{x}) < f(x), \quad \forall x \in (\Omega \cap N_\varepsilon(\bar{x})) \setminus \{\bar{x}\}.$$

- Global minimum:

$$\exists \varepsilon > 0, \text{ such that } f(\bar{x}) \leq f(x), \quad \forall x \in \Omega.$$

- Strict global minimum:

$$\exists \varepsilon > 0, \text{ such that } f(\bar{x}) < f(x), \quad \forall x \in \Omega \setminus \{\bar{x}\}.$$

Topology Concepts

- The interior of Ω :

$$\text{Int}(\Omega) = \{x \in \Omega : N_\varepsilon(x) \subseteq \Omega \text{ for some } \varepsilon > 0\}.$$

- Ω is open if $\Omega = \text{Int}(\Omega)$, i.e., every point of Ω is an interior point

- The exterior of Ω :

$$\text{Ext}(\Omega) = \{x : N_\varepsilon(x) \subseteq E^n \setminus \Omega \text{ for some } \varepsilon > 0\}$$

where

$$E^n = \{x = (x_1, \dots, x_n) : x_i \in \mathbb{R}\},$$
$$E^n \setminus \Omega = \{x \in E^n : x \notin \Omega\}.$$



Topology Concepts

- x is a boundary point of Ω :

$$\exists x_1 \in \Omega : x_1 \in N_\varepsilon(x), \quad \forall \varepsilon > 0,$$

$$\exists x_2 \notin \Omega : x_2 \in N_\varepsilon(x), \quad \forall \varepsilon > 0.$$

- The boundary $\text{Bd}(\Omega)$:

The collection of all boundary points of Ω

- A set Ω is closed:

$$\text{Bd}(\Omega) \subseteq \Omega$$

- x is an accumulation point of Ω :

$$\exists x' \neq x, \quad x' \in \Omega : \quad x' \in N_\varepsilon(x), \quad \forall \varepsilon > 0.$$



Post-Optimal (Sensitivity) Analysis

- Data in a mathematical programming problem can change as conditions change: Data
→ Estimates/Measurements
- Possible to solve new problem by modifying the original one, thus avoiding completely solving the new problem
- Post-optimal analysis:

optimal solution of
existing problem



optimal solution of
modified problem



Example 5

- A company's products are stored in m places, and sold in n outlets
- Shipping cost from the i -th storage to the j -th outlet: $c_{i,j}$ dollars
- One day, each outlet requires exactly d_j units of product
- Each storage has s_i units of product available for shipment
- Question: To minimize the shipping cost, determine the number of units should be shipped from each storage to each outlet



Example 5

- Question: To minimize the shipping cost, determine the number of units should be shipped from each storage to each outlet

$$\min_{x_{i,j}} \sum_{i=1}^m \sum_{j=1}^n c_{i,j} x_{i,j}$$

subject to

$$\sum_{j=1}^n x_{i,j} \leq s_i, \quad \forall i,$$

$$\sum_{i=1}^m x_{i,j} = d_j, \quad \forall j,$$

$$x_{i,j} \geq 0, \quad \forall i, j.$$

Parametric Programming Problem

- Question: Product is good. Customers want more. Outlet requires an increase of $\bar{d}_j$ per day, and after t days

$$\min_{x_{i,j}} \sum_{i=1}^m \sum_{j=1}^n c_{i,j} x_{i,j}$$

subject to

$$\sum_{j=1}^n x_{i,j} \leq s_i, \quad \forall i,$$

$$\sum_{i=1}^m x_{i,j} = d_j + t\bar{d}_j, \quad \forall j,$$

$$x_{i,j} \geq 0, \quad \forall i, j,$$

$$t \geq 0.$$



Model Stability

- **Model Stable:** small changes in the data $\rightarrow$ small changes in $x_1^*, \dots, x_n^*$ and $f(x_1^*, \dots, x_n^*)$
- **Model Unstable:** small changes in the data $\rightarrow$ LARGE changes in $x_1^*, \dots, x_n^*$ and $f(x_1^*, \dots, x_n^*)$
- Frequently, model-unstable problems possess redundant constraints

$$g_i(x_1, \dots, x_n) = b_i, \quad \forall i, \\ \sum_{i=1}^m \lambda_i g_i(x_1, \dots, x_n) = \sum_{i=1}^m \lambda_i b_i$$

Feasible problem

$\Rightarrow$

$$g_i(x_1, \dots, x_n) = b_i, \quad \forall i, \\ \sum_{i=1}^m \lambda_i g_i(x_1, \dots, x_n) = \varepsilon + \sum_{i=1}^m \lambda_i b_i$$

Infeasible problem

A Review of Linear Algebra: Matrix

■ Matrix:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

■ Compact expression: $A = [a_{i,j}]$

- $a_{i,j}$: the (i,j) -th entry of A
- $\mathbb{R}^{m \times n}$: The collection of all $m \times n$ real matrices



A Review of Linear Algebra: Matrix

- The transpose of A :

$$A^T = \begin{bmatrix} a_{1,1} & a_{2,1} & \cdots & a_{m,1} \\ a_{1,2} & a_{2,2} & \cdots & a_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & \cdots & a_{m,n} \end{bmatrix}$$



A Review of Linear Algebra: Vector

- Column matrix (vector):

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = [x_1, \dots, x_m]^T \in \mathbb{R}^{m \times 1}$$

- Row matrix:

$$y^T = [y_1, \dots, y_n] \in \mathbb{R}^{1 \times n}$$

- Scalars: real/complex numbers
- Vector space ($\mathbb{R}^{m \times 1}$): Euclidean m -space (E^m)



A Review of Linear Algebra: Vector

- Vector addition:

$$x + y = [x_1 + y_1, \dots, x_n + y_n]^T, \text{quad } x, y \in \mathbb{R}^{n \times 1}$$

- For a scalar α , scalar multiplication:

$$\alpha x = [\alpha x_1, \dots, \alpha x_n]^T$$



A Review of Linear Algebra: Vector

■ Inner product:

$$x^T y = \langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

■ Properties:

- $\langle x, y \rangle = \langle y, x \rangle$
- $\langle (\alpha x + \beta y), z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$
- $\langle x, x \rangle \geq 0$, with equality holds only when $x_i = 0, \forall i$
- Euclidean norm: $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x^T x}$



A Review of Linear Algebra: Subspace

- **Subspace** S of $\mathbb{R}^{n \times 1}$: Any **nonempty** set of vectors, such that $x, y \in S$, $\alpha, \beta \in \mathbb{R}$, and $\alpha x + \beta y \in S$
- Linear combination:

$$\begin{aligned}\sum_{i=1}^k \alpha_i x_i &= \alpha_1 x_1 + \dots + \alpha_k x_k \\ &= \left[\sum_{i=1}^k \alpha_i x_{i,1}, \dots, \sum_{i=1}^k \alpha_i x_{i,n} \right]^T\end{aligned}$$

- If $B \subseteq \mathbb{R}^{n \times 1}$, all linear combination of elements of B :

$$L(B) \triangleq \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \in \mathbb{R}, x_i \in B, k \in \mathbb{N} \right\}$$



Example 5

Let $B_1 = \{[1, 0, 0]^T, [0, 1, 0]^T, [0, 0, 1]^T\}$.

$$\begin{aligned} L(B_1) &= \{x_1[1, 0, 0]^T + x_2[0, 1, 0]^T + x_3[0, 0, 1]^T : x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \{[x_1, x_2, x_3]^T : x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \mathbb{R}^{3 \times 1}. \end{aligned}$$

Example 6

Let $B_2 = \{[1, 0, 0]^T, [1, 0, 1]^T\}$.

$$\begin{aligned} L(B_2) &= \{x_1[1, 0, 0]^T + x_2[1, 0, 1]^T : x_1, x_2 \in \mathbb{R}\} \\ &= \{[x_1 + x_2, 0, x_2]^T : x_1, x_2 \in \mathbb{R}\}. \end{aligned}$$

$L(B_2)$ is the x_1 - x_3 plane.



Example 6

Let $B_3 = \{[1, 0, 0]^T, [1, 0, 1]^T, [4, 0, 1]^T\}$.

$$\begin{aligned} L(B_3) &= \{x_1[1, 0, 0]^T + x_2[1, 0, 1]^T + x_3[4, 0, 1]^T : x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \{[x_1 + x_2 + 4x_3, 0, x_2 + x_3]^T : x_1, x_2, x_3 \in \mathbb{R}\}. \end{aligned}$$

$L(B_3) = L(B_2)$, i.e., the x_1 - x_3 plane, again.



A Review of Linear Algebra: Subspace

- $L(B)$ is a subspace of $\mathbb{R}^{n \times 1}$: $L(B)$ is a subspace spanned (or generated) by B
- If $S = L(B)$, then B spans S
- ◇ Any intersection of subspaces of $\mathbb{R}^{n \times 1}$ is also a subspace of $\mathbb{R}^{n \times 1}$
- ◇ $L(B)$ is the intersection of all subspaces of $\mathbb{R}^{n \times 1}$ that contain $B \rightarrow L(B)$ is the “smallest” subspace that contains B



Example 7

Let $S = \{[x_1, x_2, x_3]^T : x_1 + x_2 = 0, x_1, x_2, x_3 \in \mathbb{R}\}$.

Question: Prove that S is a subspace of $\mathbb{R}^{3 \times 1}$, and find a set that spans S

Solution:

$$\begin{aligned} S &= \{[x_1, x_2, x_3]^T : x_1 + x_2 = 0, x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \{[x_1, -x_1, x_3]^T : x_1, x_3 \in \mathbb{R}\} \\ &= \{x_1[1, -1, 0]^T + x_3[0, 0, 1]^T : x_1, x_3 \in \mathbb{R}\} \\ &= L(\{[1, -1, 0]^T, [0, 0, 1]^T\}). \end{aligned}$$

S is a subspace that is spanned by $\{[1, -1, 0]^T, [0, 0, 1]^T\}$.



A Review of Linear Algebra: Subspace

■ Definition:

$B \subseteq \mathbb{R}^{n \times 1}$, B is a **linearly dependent** set of vectors

$$\iff$$

$\exists$ distinct $x_1, \dots, x_k \in B$, $\alpha_1, \dots, \alpha_k \in \mathbb{R}$, not ALL $\alpha_i = 0$, such that

$$\alpha_1 x_1 + \dots + \alpha_k x_k = 0.$$

0 is the vector having all zero entries, i.e., $0 = [0, \dots, 0]^T$

■ B is a **basis** for S if:

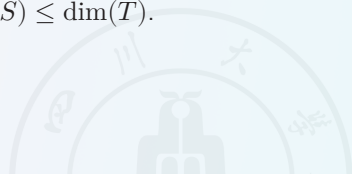
1 $L(B) = S$, i.e., B spans S

2 B is linearly independent



A Review of Linear Algebra: Subspace

- Theorem 1: Every subspace, S of $\mathbb{R}^{n \times 1}$, has a basis. Any two bases of a subspace must contain the same number of elements ([Dimension](#)).
- Theorem 2: Every set of vectors that spans a subspace contains a basis for the subspace.
- Theorem 3: Any linearly independent subset of a subspace can be extended to a basis for the space.
- Theorem 4: If S is a subspace of $\mathbb{R}^{n \times 1}$ with $\dim(S) = k$, then any subset of S that contains more than k vectors is a linearly dependent set of vectors.
 - If S and T are both subspace of $\mathbb{R}^{n \times 1}$, and $S \subseteq T$, then $\dim(S) \leq \dim(T)$.



A Review of Linear Algebra: Subspace

- An $m \times n$ real matrix A :

$$A = [a_{i,j}] = \begin{bmatrix} a_{1,1} & \cdots & a_{1,j} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots & & \vdots \\ a_{i,1} & \cdots & a_{i,j} & \cdots & a_{i,n} \\ \vdots & & \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,j} & \cdots & a_{m,n} \end{bmatrix}$$

- The j -th column of A : $A^{(j)} = [a_{1,j}, \dots, a_{i,j}, \dots, a_{m,j}]^T$
- The i -th row of A : $A_{(i)} = [a_{i,1}, \dots, a_{i,j}, \dots, a_{i,n}]$



A Review of Linear Algebra: Subspace

- **Column space:** The space spanned by the columns of A , which is a subspace of $\mathbb{R}^{m \times 1}$
- **Row space:** The space spanned by the rows of A , which is a subspace of $\mathbb{R}^{n \times 1}$
- **Rank:** The dimension of the column (row) space of A is called the column (row) rank of A :

$$\text{rank}(A) = \text{rank}_{\text{col}}(A) = \text{rank}_{\text{row}}(A)$$



A Review of Linear Algebra: Matrix Operation

- Matrix addition:

$$A + B = [a_{i,j} + b_{i,j}]$$

- Matrix scaling:

$$\alpha A = [\alpha a_{i,j}]$$

- Matrix product: $A = [a_{i,j}] \in \mathbb{R}^{m \times k}$, $B = [b_{i,j}] \in \mathbb{R}^{k \times n}$

$$AB = \left[\sum_{l=1}^k a_{il} b_{lj} \right] \in \mathbb{R}^{m \times n}$$



A Review of Linear Algebra: Identity Matrix

■ Identity Matrix:

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = [\delta_{i,j}] \in \mathbb{R}^{n \times n}$$

■ Kronecker delta:

$$\delta_{i,j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$



A Review of Linear Algebra: Submatrix

■ B is a submatrix of A :

$$A = \begin{bmatrix} 6 & 1 & 3 & 4 & 0 & 5 \\ 2 & 1 & 4 & 0 & 0 & 1 \\ 1 & 0 & 1 & -1 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

A Review of Linear Algebra: Blocks

- Blocks: The submatrices in a partitioned matrix

$$A = \begin{bmatrix} 6 & 1 & 3 & 4 & 0 & 5 \\ 2 & 1 & 4 & 0 & 0 & 1 \\ 1 & 0 & 1 & -1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} B & D & O & b \\ -c_B^T & -c_D^T & 1 & 0 \end{bmatrix},$$

$$B = \begin{bmatrix} 6 & 1 \\ 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 4 \\ 4 & 0 \end{bmatrix}, \quad O = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 1 \end{bmatrix},$$
$$-c_B^T = [1 \ 0], \quad -c_D^T = [1 \ -1].$$

A Review of Linear Algebra: Blocks

- Matrix multiplication using blocks as elements

$$A = \begin{bmatrix} C & D \\ E & F \end{bmatrix}, \quad B = \begin{bmatrix} G & H & J \\ K & L & M \end{bmatrix}$$

$$AB = \begin{bmatrix} CG + DK & CH + DL & CJ + DM \\ EG + FK & EH + FL & EJ + FM \end{bmatrix}.$$



A Review of Linear Algebra: Matrix Inverse

- Two square matrices:

$$A = [a_{i,j}] \in \mathbb{R}^{n \times n}, \quad B = [b_{i,j}] \in \mathbb{R}^{n \times n}$$

- $AB = BA = I_n$:
 A is nonsingular, $B = A^{-1}$ is the inverse of A
- A NOT invertible is singular.



A Review of Linear Algebra: Matrix Inverse

- $A = [a_{i,j}] \in \mathbb{R}^{n \times n}$, A is nonsingular:

$$\text{rank}(A) = n \iff \text{columns (rows) are linearly independent}$$

- Columns are linearly dependent, so singular:

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

- Columns are linearly independent, so nonsingular:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$



A Review of Linear Algebra: Elementary Row Operations

- Type 1: $\alpha A_{(i)}$, $\alpha \neq 0$
- Type 2: $A_{(i)} + \alpha A_{(j)} \rightarrow A_{(i)}$
- Type 3: $A_{(i)} \rightleftharpoons A_{(j)}$
- Elementary row operations do not change the row space of a matrix

$$[A \mid I_n] \longrightarrow \cdots \longrightarrow [I_n \mid A^{-1}]$$

- If the above is impossible, then A is singular



A Review of Linear Algebra: Elementary Row Operations

■ Nonsingular $B \in \mathbb{R}^{m \times m}$, $-c_B^\top \in \mathbb{R}^{1 \times m}$, $0 = [0, \dots, 0]^\top$

$$\begin{bmatrix} B^{-1} & 0 \\ c_B^\top B^{-1} & 1 \end{bmatrix} \begin{bmatrix} B & 0 \\ -c_B^\top & 1 \end{bmatrix} = \begin{bmatrix} I_m & 0 \\ 0^\top & 1 \end{bmatrix} = I_{m+1}.$$

$$\Rightarrow \begin{bmatrix} B & 0 \\ -c_B^\top & 1 \end{bmatrix}^{-1} = \begin{bmatrix} B^{-1} & 0 \\ c_B^\top B^{-1} & 1 \end{bmatrix}, \quad \text{i.e., nonsingular.}$$



A Review of Linear Algebra: Elementary Row Operations

- With a specific sequence of elementary row operations

$$[B \mid I_m] \rightarrow [I_m \mid B^{-1}]$$

- With the same sequence of elementary row operations

$$\left[\begin{array}{cc|cc} B & 0 & I_m & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} I_m & 0 & B^{-1} & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right]$$

- Replace by “the last row” $+c_B^T \cdot$ “the block row”

$$\left[\begin{array}{cc|cc} I_m & 0 & B^{-1} & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} I_m & 0 & B^{-1} & 0 \\ 0^T & 1 & c_B^T B^{-1} & 1 \end{array} \right].$$



Example 8

■ Question: Solve $Ax = b$ for x where

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

Example 8

■ Solution:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 & 1 \\ -2 & 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \right]$$

$$\xrightarrow{A_{(2)}+2A_{(1)}, A_{(3)}-A_{(1)}} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 & 1 \\ 0 & 3 & -1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 2 & -1 & 0 & 1 & -1 \end{array} \right]$$

$$\xrightarrow{A_{(1)}-\frac{1}{3}A_{(2)}, \frac{1}{3}A_{(2)}} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{2}{3} & \frac{1}{3} & -\frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} & 0 & \frac{4}{3} \\ 0 & 0 & 2 & -1 & 0 & 1 & -1 \end{array} \right]$$

$$\xrightarrow{A_{(1)}+\frac{1}{3}A_{(3)}, A_{(2)}+\frac{1}{6}A_{(3)}, \frac{1}{2}A_{(3)}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -\frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{3} & \frac{1}{6} & \frac{7}{6} \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & -\frac{1}{2} \end{array} \right].$$

$$A^{-1} = \begin{bmatrix} 0 & -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}, \quad x = \begin{bmatrix} -\frac{2}{3} \\ \frac{7}{6} \\ -\frac{1}{2} \end{bmatrix}.$$

A Review of Linear Algebra: Elementary Row Operations

- Augmented matrix: $[A \mid b]$
- Gauss-Jordan method of elimination: Solving $Ax = b$ by elementary row operations, to obtain $[I_m \mid A^{-1}b]$
- Elementary matrix: Performing the elementary row operations on an identity matrix

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

A Review of Linear Algebra: Gaussian Elimination

- **Gaussian elimination:** Applying forward elimination and back substitution together to solve $Ax = b$
 - **Upper triangular matrix:** U , all zeros below the main diagonal
 - **Forward elimination:** Using elementary row operations on $[A \mid b]$ to obtain $[U \mid \bar{b}]$
 - **Back substitution:** Substituting $x_m, x_{m-1}, \dots, x_{m-i}$ into the $(m-i-1)$ -th equation and solving for x_{m-i-1}
- **Question:** Solve linear system with Gaussian elimination

$$2x_2 - x_3 = 8,$$

$$x_1 + 2x_2 = 6,$$

$$3x_2 + 4x_3 = 12.$$

A Review of Linear Algebra: Gaussian Elimination

- Solution: forward elimination on the augmented matrix $[A \mid b]$

$$\begin{bmatrix} 0 & 2 & -1 & 8 \\ 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \end{bmatrix} \xrightarrow{A_{(1)} \leftrightarrow A_{(2)}} \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 2 & -1 & 8 \\ 0 & 3 & 4 & 12 \end{bmatrix}$$

$$\xrightarrow{A_{(2)} \leftrightarrow A_{(3)}} \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \\ 0 & 2 & -1 & 8 \end{bmatrix} \xrightarrow{A_{(3)} - \frac{2}{3}A_{(2)}} \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \\ 0 & 0 & -\frac{11}{3} & 0 \end{bmatrix}.$$

By back substitution, $x_3 = 0$, $x_2 = 4$, and $x_1 = -2$.



A Review of Linear Algebra: Gaussian Elimination

■ The upper triangular matrix

$$U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & -\frac{11}{3} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}}_{F_3} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{F_2} \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{F_1} A.$$

$$F_2 F_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

A Review of Linear Algebra: LU-decomposition

- **Row echelon form:** Upper triangular (for square matrix)
 - **Pivot:** The leftmost nonzero entry of a row
 - **Pivoting:** Elementary row operations to find pivot
 - **Pivot matrix:** Matrix that performs a row interchange; it is a permutation matrix obtained by applying the same interchange to I_m
- **Permutation matrix:** the rows of I_m are permuted



A Review of Linear Algebra: LU-decomposition

- Apply Gaussian elimination to $Ax = b$ for row echelon form:

$$Ux = (F_k P_k) \cdots (F_1 P_1) Ax = (F_k P_k) \cdots (F_1 P_1) b = \bar{b}.$$

- Collecting the row permutations into a permutation matrix P and the elimination operations into a lower-triangular matrix $L = (F_k P_k)^{-1} \cdots (F_1 P_1)^{-1} = P_1^{-1} F_1^{-1} \cdots P_k^{-1} F_k^{-1}$ gives

$$PA = LU.$$

- Hence

$$Pb = PAx = LUx = Ly,$$

followed by $Ux = y$.



Affine & Convex Sets

■ For $x_i \in \mathbb{R}^{n \times 1}$, $\forall i = 1, \dots, k$

■ Linear combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \forall \alpha_i \in \mathbb{R}$$

■ Affine combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \sum_{i=1}^k \alpha_i = 1$$

■ Conical (nonnegative) combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \forall \alpha_i \geq 0$$

■ Convex combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \forall \alpha_i \geq 0 \cap \sum_{i=1}^k \alpha_i = 1$$

Affine & Convex Sets

■ $S \subseteq \mathbb{R}^{n \times 1}, S \neq \emptyset$

■ Linear hull of S :

$$L(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \in \mathbb{R}, x_i \in S, k \in \mathbb{N} \right\}$$

■ Affine hull of S :

$$A(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \in \mathbb{R}, \sum_{i=1}^k \alpha_i = 1, x_i \in S, k \in \mathbb{N} \right\}$$

Affine & Convex Sets

- Conical hull of S :

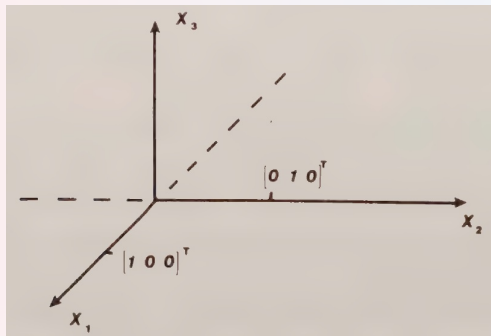
$$C(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \geq 0, x_i \in S, k \in \mathbb{N} \right\}$$

- Convex hull of S :

$$X(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \geq 0, \sum_{i=1}^k \alpha_i = 1, x_i \in S, k \in \mathbb{N} \right\}$$

Example 9

- Question: $S = \{[1, 0, 0]^T, [0, 1, 0]^T\}$, determine $L(S)$, $A(S)$, $C(S)$, and $X(S)$
- Solution:



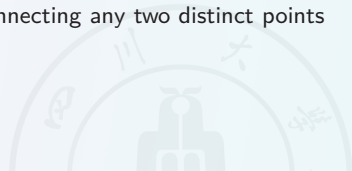
Affine & Convex Sets

- Proposition 1: For any nonempty set $S \subseteq \mathbb{R}^{n \times 1}$,

$$S \subseteq X(S) \subseteq A(S) \subseteq L(S),$$

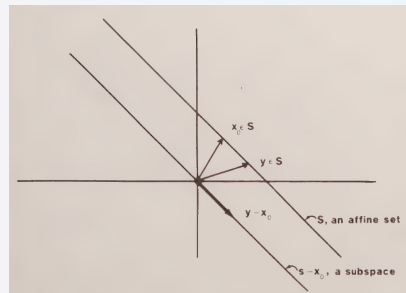
$$S \subseteq X(S) \subseteq C(S) \subseteq L(S).$$

- Proposition 2: S is a subspace of $\mathbb{R}^{n \times 1} \iff S = L(S)$
- Proposition 3: S is an affine set $\iff S = A(S) \iff$ the line passing through any two distinct points in S is a subset of S
- Proposition 4: S is a convex cone $\iff S = C(S) \iff$ every nonnegative linear combination of points in S belongs to S
- Proposition 5: S is a convex set $\iff S = X(S) \iff$ the line segment connecting any two distinct points in S is a subset of S



Affine & Convex Sets

- Proposition 6: A subset $S \subseteq \mathbb{R}^{n \times 1}$ is affine $\iff S - x_0$ is a subspace of $\mathbb{R}^{n \times 1}$, for every $x_0 \in S$
- An affine set is just the translate of a subspace
- $\dim(S) = \dim(S - x_0)$
- For any subset $T \subseteq \mathbb{R}^{n \times 1}$,
 $\dim(T) = \dim(A(T))$



Affine & Convex Sets

For $c \neq 0$ and $\beta \in \mathbb{R}$:

■ Hyperplane:

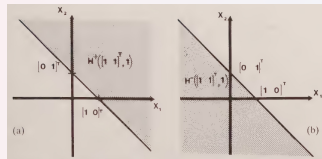
$$H(c, \beta) = \{x : c^T x = \beta\}$$

■ Upper half space:

$$H^+(c, \beta) = \{x : c^T x \geq \beta\}$$

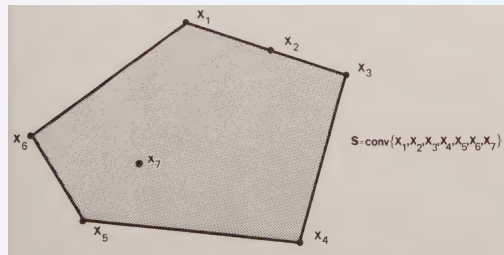
■ Lower half space:

$$H^-(c, \beta) = \{x : c^T x \leq \beta\}$$



Affine & Convex Sets

- Convex Polytope: $X(\{x_1, x_2, \dots, x_k\})$, $k < +\infty$
- Affinely Independent: $\{x_0, x_1, \dots, x_k\}$ is affinely independent $\iff \{x_1 - x_0, x_2 - x_0, \dots, x_k - x_0\}$ is linearly independent
- k -dimensional simplex: A convex polytope which is a convex hull of $k + 1$ affinely independent vectors



Affine & Convex Sets

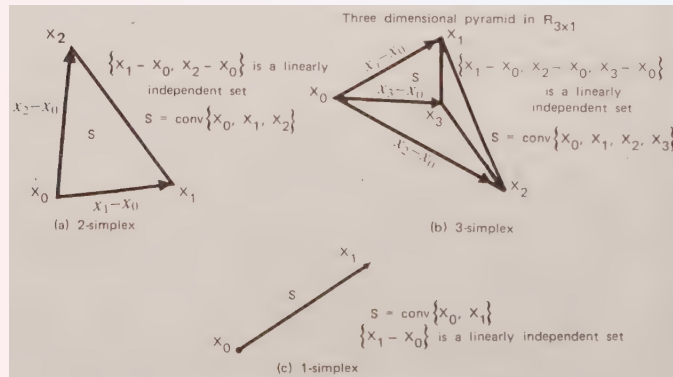
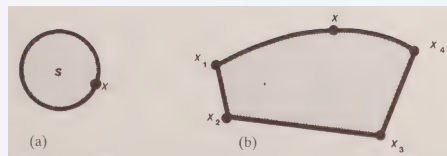


Figure 1: (a) 2-simplex, (b) 3-simplex, (c) 1-simplex

Affine & Convex Sets

- Extreme Point: $z \in S$ (convex set) is an extreme point of $S \iff$ there does NOT exist distinct $x, y \in S$ and $0 < \alpha < 1$ such that $z = \alpha x + (1 - \alpha)y$
- The vertices of any simplex are extreme points of the simplex
- The extreme points of a convex set S are those NOT on the interior of any line segment connecting any other pair of points of S



Linear Programming Problem

- Original expression

$$\min c_1x_1 + \dots + c_nx_n$$

subject to

$$a_{1,1}x_1 + \dots + a_{1,n}x_n \leq b_1,$$

$$\vdots$$

$$a_{m,1}x_1 + \dots + a_{m,n}x_n \leq b_m,$$

and

$$\forall x_i \geq 0.$$



Linear Programming Problem

- Equivalent expression:

$$\min c^T x$$

subject to

$$\begin{aligned} Ax &= A^{(1)}x_1 + A^{(2)}x_2 + \dots + A^{(n)}x_n \leq b, \\ x &\geq 0. \end{aligned}$$

- Set of feasible solutions: $F = \{x : Ax \leq b \cap x \geq 0\}$

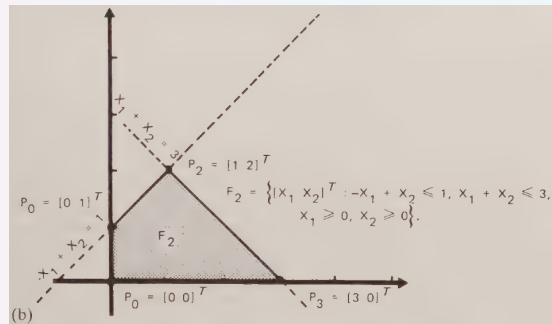
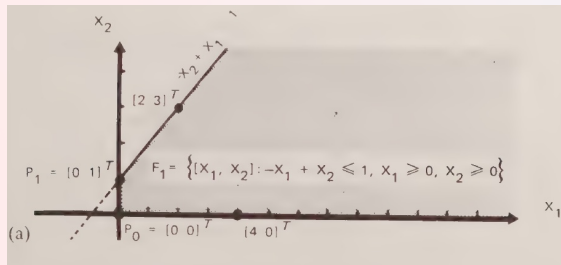


Linear Programming Problem

- Proposition 7: The set of feasible solution, F , to the linear programming problem is a convex set
- Proof: Let $x_1, x_2 \in F$, $\alpha \in [0, 1]$, check the convex combination $z = \alpha x_1 + (1 - \alpha)x_2$. Does $z \in F$?
- $F = \left(\bigcap_{i=1}^m \{x : A_{(i)}x \leq b_i\} \right) \cap \left(\bigcap_{i=1}^n \{x : x_i \geq 0\} \right)$
- Convex polyhedron: Intersection of finite number of half spaces
- Convex polytope $\implies$ Convex polyhedron; the converse does not hold in general



Linear Programming Problem



Linear Programming Problem

Finite Basis Theorem

Let $F = \{x \in \mathbb{R}^{n \times 1} : Ax \leq b\}$ be a nonempty convex polyhedron. Then there exist finite sets $\{x_1, \dots, x_d\}$ and $\{y_1, \dots, y_\ell\}$ such that

$$F = X(\{x_1, \dots, x_d\}) + C(\{y_1, \dots, y_\ell\}),$$

i.e.,

$$F = \left\{ \sum_{i=1}^d \alpha_i x_i + \sum_{j=1}^{\ell} \beta_j y_j : \alpha_i \geq 0, \beta_j \geq 0, \sum_{i=1}^d \alpha_i = 1 \right\}.$$

Linear Programming Problem

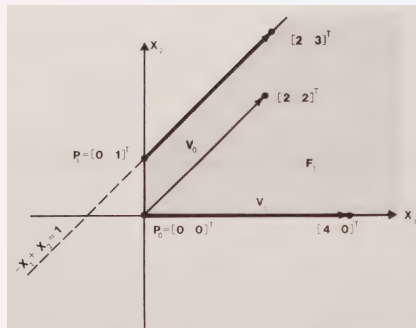


Figure 3: $F_2 = X(\{p_0, p_1\}) + C(\{v_0, v_1\})$, i.e., F_2 can be expressed as $x = \alpha_0 p_0 + \alpha_1 p_1 + \beta_0 v_0 + \beta_1 v_1$, where $\alpha_0 \geq 0$, $\alpha_1 \geq 0$, $\beta_0 \geq 0$, $\beta_1 \geq 0$ and $\alpha_0 + \alpha_1 = 1$

Theorem on Bounded Linear Functions

Theorem

If linear function $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$ is bounded from below/above (i.e., $f(\mathbf{x}) \geq \mu$) over the convex polyhedron $F = \{\mathbf{x} : \mathbf{Ax} \leq \mathbf{b}\}$, then $f(\mathbf{x})$ obtains a minimum/maximum over F , which occurs at one of the extreme points of F

Proof: If $\mathbf{x} \in F$, then $\mathbf{x} = \sum_{i=1}^d \alpha_i \mathbf{x}_i + \sum_{i=1}^{\ell} \beta_i \mathbf{y}_i$, where $\alpha_i \geq 0$ $\beta_i \geq 0$ $\sum_{i=1}^d \alpha_i = 1$. Let $f(\mathbf{x}_k) = \min\{f(\mathbf{x}_1), \dots, f(\mathbf{x}_d)\}$

$$f(\mathbf{x}) = f\left(\sum_{i=1}^d \alpha_i \mathbf{x}_i + \sum_{i=1}^{\ell} \beta_i \mathbf{y}_i\right) = \sum_{i=1}^d \alpha_i f(\mathbf{x}_i) + \sum_{i=1}^{\ell} \beta_i f(\mathbf{y}_i) \geq \sum_{i=1}^d \alpha_i f(\mathbf{x}_i) \geq \sum_{i=1}^d \alpha_i f(\mathbf{x}_k) = f(\mathbf{x}_k)$$

Quiz: Minimize Linear Function

- Minimize $f(x_1, x_2) = -x_1 + x_2$, subject to the constraints

$$-x_1 + 2x_2 \leq 10$$

$$5x_1 + 3x_2 \leq 41$$

$$2x_1 - 3x_2 \leq 8$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Quiz: Matrix Notation

- Minimize $f(x) = c^T x$, subject to $Ax \leq b$, $x \geq 0$

Note that $\mathbf{x} = [x_1, x_2]^T$, $\mathbf{c} = [-1, 1]^T$, $\mathbf{b} = [10, 41, 8]^T$

[

]

LP Problem: Slack Variables

Finding the optimal solution to the LPP $\rightarrow$ Finding extreme points of the convex polyhedron of F

- Change inequality constraints to equality constraints via slack variables

$$a_{1,1}x_1 + \dots + a_{1,n}x_n + x_{n+1} = b_1$$

$$a_{2,1}x_1 + \dots + a_{2,n}x_n + x_{n+2} = b_2$$

$$\vdots$$

$$a_{m,1}x_1 + \dots + a_{m,n}x_n + x_{n+m} = b_m$$

where $x_i \geq 0, \forall i \in \{1, 2, \dots, n + m\}$



LP Problem: Standard Form

1. $\min \hat{c}^T \hat{x}$, $\hat{A}\hat{x} = \mathbf{b}$, $\hat{x} \geq 0$, where $\hat{c} = [c_1, \dots, c_n, 0, \dots, 0]^T$
2. Feasible set: $\hat{F} = \{\hat{x} \in \mathbf{R}_{(n+m) \times 1} : \hat{A}\hat{x} = \mathbf{b}, \hat{x} \geq 0\}$
3. Coefficient matrix

$$\hat{A} = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} & 1 & 0 & \dots & 0 \\ a_{2,1} & \dots & a_{2,n} & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & \dots & a_{m,n} & 0 & 0 & \dots & 1 \end{bmatrix} = [\mathbf{A} | \mathbf{I}_m]$$

Mapping between F and $\hat{F}$

- Define a function $\sigma : F \rightarrow \hat{F}$, i.e.,

$$\sigma(\mathbf{x}) = \hat{\mathbf{x}} = \begin{bmatrix} \mathbf{x} \\ \mathbf{b} - \mathbf{Ax} \end{bmatrix}$$

- Proposition 8: $\sigma(\cdot)$ is a one-to-one mapping that preserves convex combinations of vectors, i.e.,

$$\sigma(\alpha x + (1 - \alpha)y) = \alpha\sigma(x) + (1 - \alpha)\sigma(y)$$

- Proposition 9: $\mathbf{x}$ is an extreme point of $F \iff \hat{\mathbf{x}}$ is an extreme point of $\hat{F}$



Example: Mapping σ

$$F = \{[x_1, x_2]^T : x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}$$

$$\hat{F} = \{[x_1, x_2, x_3]^T : x_1 + x_2 + x_3 = 1, x_1 \geq 0, x_2 \geq 0, x_3 \geq 0\}$$

$$\sigma([1, 0]^T) = [1, 0, 0]^T, \quad \sigma([0, 1]^T) = [0, 1, 0]^T, \quad \sigma([0, 0]^T) = [0, 0, 1]^T$$

Theorem on Extreme Points of $\hat{F}$

Theorem

If $\hat{x}$ is an extreme point of $\hat{F}$ and $\Delta = \{j : |\hat{x}_j| > 0\}$, then the columns $\{\hat{A}^{(j)} : j \in \Delta\}$ is linearly independent

If

- (a) $\{\hat{A}^{(i)} : i \in \Delta\}$ is some collection of m linearly independent columns of $\hat{A}$
- (b) $\mathbf{B}$ is the matrix whose columns are $\{\hat{A}^{(i)} : i \in \Delta\}$
- (c) $\mathbf{x}^*$ is a solution to $\mathbf{B}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$

Then the vector $\hat{x}$ whose i -th entry

$$[\hat{x}]_i = \begin{cases} 0, & i \notin \Delta \\ [\mathbf{x}^*]_i, & i \in \Delta \end{cases}$$

is an extreme point of $\hat{F}$



Example: Reformulation with Slack

- Minimize $f(x_1, x_2) = -x_1 + x_2$, subject to the constraints

$$-x_1 + 2x_2 \leq 10$$

$$5x_1 + 3x_2 \leq 41$$

$$2x_1 - 3x_2 \leq 8$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

- Reformulation: $\min\{-x_1 + x_2 + 0 \cdot x_3 + 0 \cdot x_4 + 0 \cdot x_5\}$, subject to $x_i \geq 0, \forall i \in \{1, \dots, 5\}$, and

$$\begin{bmatrix} -1 & 2 & 1 & 0 & 0 \\ 5 & 3 & 0 & 1 & 0 \\ 2 & -3 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 10 \\ 41 \\ 8 \end{bmatrix}$$

Example: Finding Extreme Points

At most $\binom{5}{3} = 10$ sets of three linearly independent columns in the coefficient matrix
Select the first three columns ($x_4 = x_5 = 0$)

$$\mathbf{B} = [\hat{\mathbf{A}}^{(1)}, \hat{\mathbf{A}}^{(2)}, \hat{\mathbf{A}}^{(3)}]$$

$x_1 = 7, x_2 = 2, x_3 = 13$ all nonnegative $\mathbf{x}^* = [7, 2, 13]^T \rightarrow \hat{\mathbf{x}} = [7, 2, 13, 0, 0]^T$ is an extreme point of $\hat{F} \rightarrow [7, 2]^T$ is an extreme point of F



Example: Invalid Basis

- Select $\hat{A}^{(1)}, \hat{A}^{(2)}, \hat{A}^{(4)}$ ($x_3 = x_5 = 0$)

$$\mathbf{B} = [\hat{\mathbf{A}}^{(1)}, \hat{\mathbf{A}}^{(2)}, \hat{\mathbf{A}}^{(4)}] = \begin{bmatrix} -1 & 2 & 0 \\ 5 & 3 & 1 \\ 2 & -3 & 0 \end{bmatrix}$$

Solving $\mathbf{B}\mathbf{x} = \mathbf{b}$ gives $x_1 = 46$, $x_2 = 28$, $x_4 = -273$ (negative), so $[46, 28, 0, -273, 0]^T \notin \hat{F}$, hence $[46, 28]^T$ is NOT an extreme point of F .



Primal Simplex: Equality Form

Any LLP can be expressed as:

$$\min \mathbf{c}^T \mathbf{x} \text{ subject to } \mathbf{Ax} = \mathbf{b} \text{ and } \mathbf{x} \geq \mathbf{0}$$

$\mathbf{A} \in \mathbf{R}_{m \times n}$ $\mathbf{x} \in \mathbf{R}_{n \times 1}$, and $\text{rank}(\mathbf{A}) = m \leq n$

All the constraints are equality constraints

Each $\mathbf{x}_i$ is restricted to be nonnegative

Equivalent problem with $z_o = \mathbf{c}^T \mathbf{x}$: Finding a solution $\mathbf{x} \in \mathbf{R}_{n \times 1}$ and $z_o \in \mathcal{R}$

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ -\mathbf{c}^T \mathbf{x} + z_o &= 0 \end{aligned}$$

for $\mathbf{x} \geq \mathbf{0}$ and z_o is minimal



Primal Simplex: Matrix Equation

Matrix equation

Assume $\text{rank}(\mathbf{A}) = m$:

$$\mathbf{A} = \underbrace{[\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(m)}, \mathbf{A}^{(m+1)}, \dots, \mathbf{A}^{(n)}]}_{:=\mathbf{B}} = \mathbf{D}$$

Initial matrix for the problem: $[\quad A, b, c, z_o \quad]$



Primal Simplex: Initial Tableau

- Left multiplying the initial matrix by: $[\quad]$ - It results simplex tableau: $[\quad]$ -
Can be alternatively done by a sequence of elementary row operations



Primal Simplex: Simplex Tableau

Above augmented matrix represents the linear system:

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{B}^{-1}\mathbf{D} & \mathbf{0} \\ \mathbf{0}^T & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_D \\ z_o \end{bmatrix} = \begin{bmatrix} \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \end{bmatrix}$$

$$\begin{cases} \mathbf{x}_B + \mathbf{B}^{-1}\mathbf{D}\mathbf{x}_D = \mathbf{B}^{-1}\mathbf{b} \\ (\mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T)\mathbf{x}_D + z_o = \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \end{cases}$$

If $\mathbf{B}^{-1}\mathbf{b} \geq 0$, $\mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T \leq 0^T$:

$$\begin{cases} z_o = \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \\ \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{x}_D = \mathbf{0} \end{cases}$$



Example: Primal Simplex

Question:

$$\min -2x_1 - x_2 + 2x_3$$

subject to

$$x_1 + x_3 = 4$$

$$-2x_1 + x_2 = 8$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

$$x_3 \geq 0$$

Example: Basis Selection

Solution: By elementary row operations, $[\quad]$ In above, $\mathbf{B} = [\mathbf{A}^{(3)}, \mathbf{A}^{(2)}]$

Theorem 1: Optimality Condition

Theorem 1: Solution for $\min \mathbf{c}^T \mathbf{x}$ subject to $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$
 If $\mathbf{B}^{-1}\mathbf{b} \geq \mathbf{0}$ and $\mathbf{c}_B \mathbf{B}^{-1} \mathbf{D} - \mathbf{c}_D^T \leq \mathbf{0}^T$, then the feasible solution

$$\mathbf{x}_o = \begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_D \end{bmatrix} := \begin{bmatrix} \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{0} \end{bmatrix}$$

is optimal, and $z_o = \min \mathbf{c}^T \mathbf{x} = \mathbf{c}^T \mathbf{x}_o = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b}$

Proof: Suppose an **arbitrary** feasible solution $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$

$$\begin{aligned} \mathbf{b} &= \mathbf{Ax} = [\mathbf{B}, \mathbf{D}]\bar{\mathbf{x}} = \mathbf{B}[\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]\bar{\mathbf{x}} = \mathbf{B}\mathbf{x}_B \\ &\Rightarrow \mathbf{B}[\mathbf{x}_B - [\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]\bar{\mathbf{x}}] = \mathbf{0} \\ &\Rightarrow \mathbf{x}_B = [\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]\bar{\mathbf{x}} \\ &\Rightarrow \mathbf{c}_B^T \mathbf{x}_B = [\mathbf{c}_B^T, \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{D}] \bar{\mathbf{x}} \\ &\leq [\mathbf{c}_B^T, \mathbf{c}_D^T] \bar{\mathbf{x}} = \mathbf{c}^T \bar{\mathbf{x}} \end{aligned}$$

$\mathbf{c}_B^T \mathbf{x}_B$ is the minimal value for the problem, and $\mathbf{x}_o$ is optimal.

Basic Solutions and BFS

- Basic solution of $\mathbf{Ax} = \mathbf{b}$:

$$\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}, \quad \mathbf{x}_D = \mathbf{0}$$

- Basis matrix: the nonsingular matrix $\mathbf{B}$
- Admissible basis: $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(m)}$
- Basic feasible solution (BFS): a basic solution that is feasible
- Nondegenerate BFS: A BFS for which $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} > \mathbf{0}$
- Theorem 1 gives a sufficient condition, i.e., $\mathbf{c}_B\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T \leq \mathbf{0}^T$, for a BFS $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} > \mathbf{0}$, and $\mathbf{x}_D = \mathbf{0}^T$ to be an optimal solution.



Feasible, Basic, and BFS

- Feasible Solution: "Qualified", wide range candidates

$$\hat{A}\hat{x} = \mathbf{b}, \quad \hat{x} \geq 0$$

- Basic Solution: "Simple & Efficient", not necessarily feasible

$$\hat{A}\hat{x} = \mathbf{b}, \quad \mathbf{x}_D = \mathbf{0}$$

- BFS: "Key candidates" for optimal

$$\hat{A}\hat{x} = \mathbf{b}, \quad \mathbf{x}_D = \mathbf{0}, \quad \hat{x} \geq 0$$



Example: LP with Slack

Question: $\min 10x_1 + 18x_2$ subject to

$$5x_1 + 2x_2 \leq 170$$

$$2x_1 + 3x_2 \leq 100$$

$$x_1 + 5x_2 \leq 150$$

$$x_i \geq 0, \forall i \in \{1, 2\}$$

Reformation:

$$5x_1 + 2x_2 + x_3 = 170$$

$$2x_1 + 3x_2 + x_4 = 100$$

$$x_1 + 5x_2 + x_5 = 150$$

$$x_i \geq 0, \forall i \in \{1, 2, \dots, 5\}$$

Example: Geometric Interpretation

$[$ $]$ (10): O, A, B, E, F, G, H, I
 (5): O, A, B, C, E
 (1): C

Simplex Tableau Details

The Primal Simplex Procedure

Simplex Tableau:

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{B}^{-1}\mathbf{D} & 0 & \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{0}^T & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T & 1 & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \end{bmatrix}$$

$$X_B \triangleq [\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]$$

$$X_B = \begin{bmatrix} 1 & 0 & \cdots & 0 & x_{1,m+1} & \cdots & x_{1,j} & \cdots & x_{1,n} \\ 0 & 1 & \cdots & 0 & x_{2,m+1} & \cdots & x_{2,j} & \cdots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & x_{m,m+1} & \cdots & x_{m,j} & \cdots & x_{m,n} \end{bmatrix}$$

$$z_j \triangleq \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{A}^{(j)}, c_j \triangleq [\mathbf{c}]_j, \forall j = 1, \dots, n$$

$$z_j - c_j = \begin{cases} 0 & j = 1, \dots, m \\ [\mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{D} - \mathbf{c}_D^T]_j & j = m + 1, \dots, n \end{cases}$$



Theorem 2: Unboundedness

Theorem 1: If $z_j - c_j \leq 0, \forall j = 1, \dots, n$, the BFS $\{x_B = \mathbf{B}^{-1}\mathbf{b} \geq 0, \mathbf{x}_D = 0\}$ is an optimal solution

What if $z_j - c_j > 0$, for some j ?

↓

Theorem 2: If there exists a j so that $z_j - c_j > 0$ and each $x_{i,j} \leq 0$, then the LLP

$$\min \mathbf{c}^T \mathbf{x} \text{ subject to } \mathbf{Ax} = \mathbf{b} \text{ and } \mathbf{x} \geq \mathbf{0}$$

has an unbounded minimum over F



Theorem 3 & 4: Pivoting and Optimality

Theorem 3

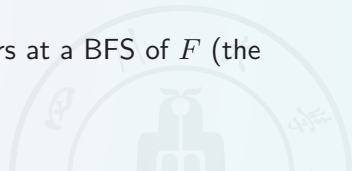
If there exists a j so that $z_j - c_j > 0$, and if some $x_{i,j} > 0$, then by pivoting at the element $x_{k,j}$, a new basic feasible solution is generated that does not increase the value of z_o . Note that k is selected such that

$$\theta \triangleq \frac{[\mathbf{B}^{-1}\mathbf{b}]_k}{x_{k,j}} = \min \left\{ \frac{[\mathbf{B}^{-1}\mathbf{b}]_i}{x_{i,j}} : x_{i,j} > 0 \right\}$$

If $\mathbf{B}^{-1}\mathbf{b} > 0$, i.e., the BFS $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}$ and $\mathbf{x}_D = \mathbf{0}$ is nondegenerate, then the value of z_o decreases.

Theorem 4

If $\mathbf{c}^T \mathbf{x}$ obtains a finite minimum over F , then the minimum occurs at a BFS of F (the minimum may occur at other places as well)



Example: Introducing Slack

Question

$$\min \mathbf{c}^T \mathbf{x}$$

subject to

$$3x_1 + x_3 - x_4 = 1$$

$$7x_1 + x_2 - x_4 = 0$$

$$-8x_1 + x_4 \leq 1$$

$$x_i \geq 0, \forall i$$



Example: Augmented Form

- Add a slack variable

$$\min \mathbf{c}^T \mathbf{x}$$

subject to []

$$\mathbf{c} = [1, -1, 2, -3, 0]^T$$

Example: Basis and BFS

Example 14

$$\begin{bmatrix} 3 & 0 & 1 & -1 & 0 & 0 & 1 \\ 7 & 1 & 0 & -1 & 0 & 0 & 0 \\ -8 & 0 & 0 & -1 & 1 & 0 & 1 \\ -1 & 1 & -2 & -3 & -1 & 0 & 1 \end{bmatrix}$$

Row operation: $\mathbf{A}_{(4)} - (\mathbf{A}_{(2)} - 2\mathbf{A}_{(1)})$

$$\begin{bmatrix} 3 & 0 & 1 & -1 & 0 & 0 & 1 \\ 7 & 1 & 0 & -1 & 0 & 0 & 0 \\ -8 & 0 & 0 & -1 & 1 & 0 & 1 \\ -2 & 0 & 0 & -2 & -1 & 1 & 2 \end{bmatrix}$$

$\mathbf{B} = [\mathbf{A}^{(3)}, \mathbf{A}^{(2)}, \mathbf{A}^{(5)}]$ is nonsingular

$\mathbf{D} = [\mathbf{A}^{(1)}, \mathbf{A}^{(4)}]$

BFS: $[x_3, x_2, x_5]^T = [1, 0, 1]^T, [x_1, x_4]^T = [0, 0]^T$ (degenerate)



Example: Pivot and Unboundedness

Example 14

$$\begin{bmatrix} 3 & 0 & 1 & -1 & 0 & 0 & 1 \\ 7 & 1 & 0 & -1 & 0 & 0 & 0 \\ -8 & 0 & 0 & -1 & 0 & 1 & 1 \\ -2 & 0 & 0 & -2 & 0 & 1 & 2 \end{bmatrix}$$

Row operations: $\mathbf{A}_{(1)} + \mathbf{A}_{(3)}$, $\mathbf{A}_{(2)} + \mathbf{A}_{(3)}$, $\mathbf{A}_{(4)} - 2\mathbf{A}_{(3)}$

$$\begin{bmatrix} -5 & 0 & 1 & 0 & 1 & 0 & 2 \\ -1 & 1 & 0 & 0 & 1 & 0 & 1 \\ -8 & 0 & 0 & -1 & 0 & 1 & 1 \\ -14 & 0 & 0 & -2 & 0 & 1 & 0 \end{bmatrix}$$

(1) Pivot in the 4th column, since $z_4 - c_4 = 2 > 0$

(2) $\min \left\{ \frac{|\mathbf{B}^{-1}\mathbf{b}_i|}{x_{i,4}} : x_{i,4} > 0 \right\} = 1 \Rightarrow k = 3 \Rightarrow \text{pivoting } x_{3,4}$

(3) $\mathbf{B} = [\mathbf{A}^{(3)}, \mathbf{A}^{(2)}, \mathbf{A}^{(4)}]$ and $\mathbf{D} = [\mathbf{A}^{(1)}, \mathbf{A}^{(5)}]$

(4) BFS: $x_3 = 2$, $x_2 = 1$, $x_4 = 1$, and $x_1 = x_5 = 0$



Example: Maximization to Minimization

Question

$$\max 2x_1 + x_2 + 3x_3$$

subject to

$$x_1 + 3x_2 + 2x_3 \leq 6$$

$$-x_1 + x_2 - 3x_3 \geq -9$$

$$x_i \geq 0, \quad \forall i$$

Transformation

$$\min -2x_1 - x_2 - 3x_3$$

subject to

$$x_1 + 3x_2 + 2x_3 + x_4 = 6$$

$$x_1 - x_2 + 3x_3 + x_5 = 9$$

$$x_i \geq 0, \quad \forall i$$



Example: Simplex Iterations

Example 15

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ -1 & 1 & -3 & 0 & 1 & 0 & 9 \end{bmatrix}$$

Row operations: $\mathbf{A}_{(2)} - \frac{3}{2}\mathbf{A}_{(1)}$, $\mathbf{A}_{(3)} - \frac{3}{2}\mathbf{A}_{(1)}$, $\frac{1}{2}\mathbf{A}_{(1)}$

$$\begin{bmatrix} \frac{1}{2} & \frac{3}{2} & 1 & \frac{1}{2} & 0 & 0 & 3 \\ -\frac{1}{2} & -\frac{7}{2} & 0 & -\frac{3}{2} & 1 & 0 & 0 \end{bmatrix}$$

BSF: $\mathbf{x} = [0, 0, 0, 6, 9]^T$, with admissible basis of $\{\mathbf{A}^{(4)}, \mathbf{A}^{(5)}\}$

Pivot in the 3rd column, since $z_3 - c_3 = 3 > 0$

$$\theta = \min \left\{ \frac{|\mathbf{B}^{-1}\mathbf{b}|_i}{x_{i,3}} : x_{i,3} > 0 \right\} = \min \left\{ \frac{6}{2}, \frac{9}{3} \right\} = 3, \text{ either pivoting } x_{1,3} \text{ or } x_{2,3}$$



Example: Degeneracy and Pivot

Example 15

$$\begin{bmatrix} \frac{1}{2} & \frac{3}{2} & 1 & \frac{1}{2} & 0 & 0 & 3 \\ -\frac{1}{2} & -\frac{7}{2} & 0 & -\frac{3}{2} & 1 & 0 & 0 \end{bmatrix}$$

Row operations: $\mathbf{A}_{(2)} + \mathbf{A}_{(1)}$, $\mathbf{A}_{(3)} - \mathbf{A}_{(1)}$, $2\mathbf{A}_{(1)}$

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ 0 & -4 & 1 & -1 & 1 & 0 & 3 \end{bmatrix}$$

New BSF: $\{\mathbf{A}^{(3)}, \mathbf{A}^{(5)}\}$, $\mathbf{x} = [0, 0, 3, 0, 0]^T$ (degenerate)

Pivot in the 1st column, since $z_1 - c_1 = \frac{1}{2} > 0$

$$\theta = \min \left\{ \frac{|\mathbf{B}^{-1}\mathbf{b}|_i}{x_{i,1}} : x_{i,1} > 0 \right\} = \min\{6\} = 6, \text{ pivoting } x_{1,1}$$



Example: Optimal Solution

Example 15

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ 0 & -4 & 1 & -1 & 1 & 0 & 3 \end{bmatrix}$$

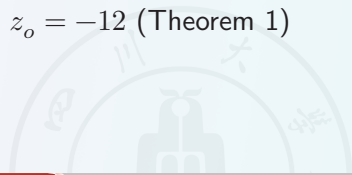
Row operation: $\mathbf{A}_{(2)} - \mathbf{A}_{(1)}$

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ 0 & -4 & -5 & -1 & -1 & 0 & -12 \end{bmatrix}$$

New BSF: $\{\mathbf{A}^{(1)}, \mathbf{A}^{(5)}\}$, $\mathbf{x} = [6, 0, 0, 0, 3]^T$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x} = [6, 0, 0, 0, 3]^T$ is the optimum, and $z_o = -12$ (Theorem 1)

Original LLP: $\max\{c^T x : x \in F\} = 12$, and $x = [6, 0, 0]^T$



Example: New Problem

Question

$$\min \mathbf{c}^T \mathbf{x}$$

subject to

$$16x_1 + 2x_2 + x_5 + 10x_6 = 4$$

$$-x_1 + x_4 + x_6 = 0$$

$$5x_1 + 4x_2 + x_3 = 8$$

$$x_i \geq 0, \quad \forall i$$

where $\mathbf{c} = [1, -1, 2, -3, 0, 0]^T$



Example: Simplex Steps

Example 16

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ -1 & -1 & -2 & -3 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\downarrow A_{(4)} + (A_{(1)} + A_{(2)} + A_{(3)})$$

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 19 & 5 & 0 & 0 & 0 & 10 & 1 & 12 \end{bmatrix}$$

BSF: $\mathbf{x} = [0, 0, 8, 0, 4, 0]^T$ (degenerate), with admissible basis of $\{\mathbf{A}^{(5)}, \mathbf{A}^{(4)}, \mathbf{A}^{(3)}\}$

Pivot in the 6th column, since $z_6 - c_6 = 10 > 0$

$\theta = \min\{0, \frac{4}{10}\} = 0$, pivoting $x_{2,6}$

Example: Pivot Selection

Example 16

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 19 & 5 & 0 & 0 & 0 & 10 & 1 & 12 \end{bmatrix}$$

$$\downarrow A_{(1)} - 10A_{(2)}, \quad A_{(4)} - 10A_{(2)}$$

$$\begin{bmatrix} 26 & 2 & 0 & -10 & 1 & 0 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 29 & 5 & 0 & -10 & 0 & 0 & 1 & 12 \end{bmatrix}$$

BSF: $\mathbf{x} = [0, 0, 8, 0, 0, 4]^T$ (degenerate), with admissible basis of $\{\mathbf{A}^{(5)}, \mathbf{A}^{(6)}, \mathbf{A}^{(3)}\}$

Pivot in the 2nd column, since $z_2 - c_2 = 5 > 0$

$\theta = \min\{\frac{4}{2}, \frac{8}{4}\} = 2$, pivoting either $x_{1,2}$ or $x_{3,2}$

Example: Further Pivoting

Example 16

$$\begin{bmatrix} 26 & 2 & 0 & -10 & 1 & 0 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 29 & 5 & 0 & -10 & 0 & 0 & 1 & 12 \end{bmatrix}$$

$$\downarrow A_{(3)} - 2A_{(1)}, \quad A_{(4)} - \frac{5}{2}A_{(1)}, \quad \frac{1}{2}A_{(1)}$$

$$\begin{bmatrix} 13 & 1 & 0 & -5 & \frac{1}{2} & 0 & 0 & 2 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -21 & 0 & 1 & 20 & -2 & 0 & 0 & 4 \\ -36 & 0 & 0 & 15 & -\frac{5}{2} & 0 & 1 & 2 \end{bmatrix}$$

BSF: $x = [0, 2, 0, 0, 0, 0]^T$ (degenerate), with admissible basis of $\{\mathbf{A}^{(2)}, \mathbf{A}^{(6)}, \mathbf{A}^{(3)}\}$

Pivot in the 4th column, since $z_4 - c_4 = 15 > 0$

$\theta = \min\{\frac{0}{1}, \frac{0}{20}\} = 0$, pivoting either $x_{2,4}$ or $x_{3,4}$

Example: Optimality Reached

Example 16

$$\begin{bmatrix} 13 & 1 & 0 & -5 & \frac{1}{2} & 0 & 0 & 2 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -21 & 0 & 1 & 20 & -2 & 0 & 0 & 4 \\ -36 & 0 & 0 & 15 & -\frac{5}{2} & 0 & 1 & 2 \end{bmatrix}$$

$$\downarrow A_{(1)} + 5A_{(2)}, \quad A_{(3)} - 20A_{(2)}, \quad A_{(4)} - 15A_{(2)}$$

$$\begin{bmatrix} 8 & 1 & 0 & 0 & \frac{1}{2} & 5 & 0 & 2 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 & -2 & -20 & 0 & 4 \\ -21 & 0 & 0 & 0 & -\frac{5}{2} & -15 & 1 & 2 \end{bmatrix}$$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x} = [0, 2, 0, 0, 0, 0]^T$ is the optimum, and $z_o = 2$ (Theorem 1)

The admissible basis: $\{\mathbf{A}^{(2)}, \mathbf{A}^{(4)}, \mathbf{A}^{(3)}\}$

Example: Multiple Optima

Question: Find several points $\mathbf{x} = [x_1, x_2, x_3, x_4]^T$ that solve the problem

$$\min 3x_1 + 2x_2 + 6x_3 + x_4$$

subject to

$$x_1 + x_4 = 4$$

$$x_1 + x_2 + x_3 = 5$$

$$x_i \geq 0, \forall i$$



Example: Finding BFS

Example 17

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 1 & 1 & 1 & 0 & 0 & 5 \\ -3 & -2 & -6 & -1 & 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 1 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix}$$

BSF: $\{\mathbf{A}^{(4)}, \mathbf{A}^{(2)}\}, \mathbf{x} = [0, 5, 0, 4]^T$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x}_1 = [0, 5, 0, 4]^T$ is the optimum, and $z_o = 14$ (Theorem 1)



Example: Alternative Optimal Solution

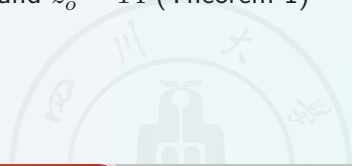
Example 17

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 1 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix} \Rightarrow \mathbf{A}_{(2)} - \mathbf{A}_{(1)} \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix}$$

Consider $z_1 - c_1 = 0$, $\theta = \min\{\frac{4}{1}, \frac{5}{1}\} = 4$, pivoting $x_{1,1}$

BSF: $\{\mathbf{A}^{(1)}, \mathbf{A}^{(2)}\}$, $\mathbf{x} = [4, 1, 0, 0]^T$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x}_2 = [4, 1, 0, 0]^T$ is also the optimum, and $z_o = 14$ (Theorem 1)



Useful Points in Simplex

Useful points when using the primal simplex procedure:

- 1 z_0 should never increase from one tableau to the next
- 2 $\mathbf{B}^{-1}\mathbf{b}$ should never contain negative components
- 3 For each $\mathbf{A}^{(j)}$ in the admissible basis, $z_j - c_j = 0$



Artificial Variables Introduction

To solve LLP with the primal simplex procedure, an initial BFS of F is required before setting up the first tableau.

This is possible after adding slack variables, $\mathbf{A} = [a_{i,j}]$ contains $\mathbf{I}_m$

Question: What if $\mathbf{A} = [a_{i,j}]$ does NOT contain $\mathbf{I}_m$?



Artificial Variables Formulation

Question:

$$\min \quad c_1x_1 + \cdots + c_nx_n + Mx_{n+1} + \cdots + Mx_{n+m}$$

subject to

$$a_{1,1}x_1 + \cdots + a_{1,n}x_n + x_{n+1} = b_1$$

$$a_{2,1}x_1 + \cdots + a_{2,n}x_n + x_{n+2} = b_2$$

$$\vdots$$

$$a_{m,1}x_1 + \cdots + a_{m,n}x_n + x_{n+m} = b_m$$

$$x_1 \geq 0, \dots, x_n \geq 0, x_{n+1} \geq 0, \dots, x_{n+m} \geq 0$$

M : an unspecified large positive number

$x_{n+1}, \dots, x_{n+m}$: artificial variables



Objective with Artificial Variables

- The objective function:

$$f(\mathbf{y}) = c_1 y_1 + \cdots + c_n y_n$$

$$\bar{f}(\bar{\mathbf{y}}) = c_1 y_1 + \cdots + c_n y_n + M y_{n+1} + \cdots + M y_{n+m}$$

- The set of feasible solutions:

$$F = \{\mathbf{y} \in \mathbf{R}_{n \times 1} : \mathbf{A}\mathbf{y} = \mathbf{b}, \mathbf{y} \geq 0\}$$

$$\bar{F} = \{\bar{\mathbf{y}} \in \mathbf{R}_{(n+m) \times 1} : [\mathbf{A}, \mathbf{I}_m]\bar{\mathbf{y}} = \mathbf{b}, \bar{\mathbf{y}} \geq 0\}$$

- Specific solution: $\begin{bmatrix} \end{bmatrix}$



Mapping between F and $\bar{F}$

$p : F \rightarrow \bar{F}$, i.e., $p(\mathbf{x}) = \bar{\mathbf{x}}$

$p(F) = \{\bar{\mathbf{x}} = p(\mathbf{x}) : \mathbf{x} \in F\}$ is a subset of $\bar{F}$

$\{f(\mathbf{x}) : \mathbf{x} \in F\} = \{\bar{f}(\bar{\mathbf{x}}) : \bar{\mathbf{x}} \in p(F)\}$

$\min\{\bar{f}(\bar{\mathbf{y}}) : \bar{\mathbf{y}} \in \bar{F}\} \leq \min\{f(\mathbf{y}) : \mathbf{y} \in F\}$

Initial Matrix with Artificials

- Objective function: $c_1x_1 + \dots + c_nx_n + Mx_{n+1} + \dots + Mx_{n+m}$
- Initial matrix:

$$\begin{bmatrix} \mathbf{A} & \mathbf{I}_m & \mathbf{0} & \mathbf{b} \\ -\mathbf{c}^T & \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & -\mathbf{1}^T & 0 & 0 \end{bmatrix}$$

- The $(m+1)$ -th row $\rightarrow c_1x_1 + \dots + c_nx_n$
- The $(m+2)$ -th row $\rightarrow Mx_{n+1} + \dots + Mx_{n+m}$



First Simplex Tableau

The 1st simplex tableau: row operations on the last row.

$$\begin{bmatrix} & M & \end{bmatrix}$$

Reduced Costs with M

- In the tableau: $[\quad M \quad]$ - To use simplex procedure, force each

$$z_j - c_j = a_j M + d_j \leq 0, \quad j = 1, \dots, n + m$$

Simplex with Artificial Variables

For $M \gg 0$ unspecified large:

to force $a_j \leq 0, \forall j$

to force $d_j \leq 0, \forall j$ if $a_j = 0$

Simple procedure involving artificial variables:

- 1 On the last row: force all $a_j^* \leq 0, j = 1, \dots, m + n$ (by iteratively selecting the column with the largest number on the last row and pivoting)
- 2 On the last-but-one row: select $\mathbf{A}^{(j)}$ with the largest positive entry over a 0 in the last row
- 3 Perform simplex procedure



Infeasibility Detection

Suppose a tableau with optimal solution:

If $a_0^* > 0$

The admissible basis contains columns that correspond to artificial variables

$F = \emptyset$: the problem is not feasible, because

$$a_0^*M + d_0^* = \min\{\bar{f}(\bar{\mathbf{y}}) : \bar{\mathbf{y}} \in \bar{F}\} \leq \min\{f(\mathbf{y}) : \mathbf{y} \in F\}$$



Optimal Solution with Artificial

Suppose a tableau with optimal solution:

If $a_0^* = 0$

The admissible basis contains columns that correspond to artificial variables

$\mathbf{x}$ is optimal solution to the original problem:

$$f(\mathbf{x}) = \bar{f}(\bar{\mathbf{x}}) = \min \{ \bar{f}(\bar{\mathbf{y}}) : \bar{\mathbf{y}} \in \bar{F} \} \leq \min \{ f(\mathbf{y}) : \mathbf{y} \in F \}$$



Example: Artificial Variables

Question:

$$\min 2x_1 + x_2 + 3x_3$$

subject to

$$x_1 - 2x_3 = 2$$

$$x_1 + x_3 = 1$$

$$x_2 + x_3 = 1$$

$$\forall x_i \geq 0$$

The coefficient matrix contains $[0, 0, 1]^T$ of $\mathbf{I}_3$. Only need 2 artificial variables.



Example: Transformation

- Transformation:

$$\min 2x_1 + x_2 + 3x_3 + Mx_4 + Mx_5$$

subject to

$$x_1 - 2x_3 + x_4 = 2$$

$$x_1 + x_3 + x_5 = 1$$

$$x_2 + x_3 = 1$$

$$\forall x_i \geq 0$$



Example: Initial Tableau

Initial matrix & the 1st simplex tableau

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ -2 & -1 & -3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(4)} + \mathbf{A}_{(3)}, \quad \mathbf{A}_{(5)} + (\mathbf{A}_{(1)} + \mathbf{A}_{(2)})$$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ -2 & -1 & -3 & 0 & 0 & 1 & 0 & 0 \\ -2 & -1 & 2 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$



Example: Infeasibility

$$[\quad \quad \quad x^* = [0, 0, 1, 2, 0]^T]$$

Example: Another Artificial Problem

Question:

$$\min 2x_1 + x_2 + x_3 + 3x_4$$

subject to

$$-\frac{1}{3}x_1 + x_3 + x_4 = 1$$

$$x_2 + x_3 = 1$$

$$-x_1 + x_2 + x_3 = 1$$

$$\forall x_i \geq 0$$

Artificial variables: x_5, x_6



Example: Transformation

- Transformation:

$$\min 2x_1 + x_2 + x_3 + 3x_4 + Mx_5 + Mx_6$$

subject to

$$-\frac{1}{3}x_1 + x_3 + x_4 = 1$$

$$x_2 + x_3 + x_5 = 1$$

$$-x_1 + x_2 + x_3 + x_6 = 1$$

$$\forall x_i \geq 0$$

Example: Initial Tableau

Example 19 Admissible basis: $\mathbf{A}^{(4)}, \mathbf{A}^{(5)}, \mathbf{A}^{(6)}$

Initial matrix & the 1st simplex tableau:

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ -2 & -1 & -1 & -3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(4)} + 3\mathbf{A}_{(1)}, \quad \mathbf{A}_{(5)} + (\mathbf{A}_{(2)} + \mathbf{A}_{(3)})$$

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ -3 & -1 & 2 & 0 & 0 & 0 & 1 & 0 & 3 \\ -1 & 2 & 2 & 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Example: Pivot

Example 19 the 1st simplex tableau & the 2nd simplex tableau:

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ -3 & -1 & 2 & 0 & 0 & 0 & 1 & 0 & 3 \\ -1 & 2 & 2 & 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(3)} - \mathbf{A}_{(2)}, \mathbf{A}_{(4)} + \mathbf{A}_{(2)}, \mathbf{A}_{(5)} - 2\mathbf{A}_{(2)}$$

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ -3 & 0 & 3 & 0 & 1 & 0 & 1 & 0 & 4 \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Example: Optimal Solution

Example 19 the 2nd simplex tableau & the 3rd simplex tableau ($a_0^* = 0$):

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ -3 & 0 & 3 & 0 & 1 & 0 & 1 & 0 & 4 \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(1)} - \mathbf{A}_{(2)}, \quad \mathbf{A}_{(4)} - 3\mathbf{A}_{(2)}$$

$$\begin{bmatrix} -\frac{1}{3} & -1 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ -3 & -3 & 0 & 0 & -2 & 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Basis contains column(s) w.r.t artificial variable(s)



Example: Artificial Variable

Question: $\min 4x_1 + x_2 + x_3 + x_4$
subject to

$$x_1 + x_4 = 4$$

$$x_1 + x_3 = 5$$

$$x_1 + x_2 + x_4 = 9$$

$$\forall x_i \geq 0$$

Artificial variables: x_5





Example: Transformation

- Transformation:

$$\min 4x_1 + x_2 + x_3 + x_4 + Mx_5$$

subject to

$$x_1 + x_4 + x_5 = 4$$

$$x_1 + x_3 = 5$$

$$x_1 + x_2 + x_4 = 9$$

$$\forall x_i \geq 0$$



Example: Initial Tableau

- Admissible basis: $\mathbf{A}^{(3)}, \mathbf{A}^{(3)}, \mathbf{A}^{(2)}$
- Initial matrix & the 1st simplex tableau:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 9 \\ -4 & -1 & -1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(5)} + \mathbf{A}_{(1)}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 9 \\ -4 & -1 & -1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Example: Optimal Solution

Example 20 the 1st simplex tableau & the 2nd simplex tableau ($a_0^* = 0$):

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 9 \\ -4 & -1 & -1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(5)} - \mathbf{A}_{(1)}, \quad \mathbf{A}_{(3)} - \mathbf{A}_{(1)}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 5 \\ -2 & 0 & 0 & 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Basis does not contain column w.r.t artificial variable



Example: Mixed Constraints

Question:

$$\min -x_1 - x_2 - 3x_3$$

subject to

$$x_1 + x_2 + x_4 = 2$$

$$x_2 + x_3 - x_4 = 1$$

$$x_1 + x_2 + x_4 \leq 5$$

Slack variables: x_4, x_5

Artificial variables: x_6, x_7





Example: Transformation

- Transformation:

$$\min -x_1 - x_2 - 3x_3 + 0x_4 + 0x_5 + Mx_6 + Mx_7$$

subject to

$$x_1 + x_2 + x_6 = 2$$

$$x_2 + x_3 - x_4 + x_7 = 1$$

$$x_1 + x_2 + x_4 + x_5 = 5$$

$$\forall x_i \geq 0$$



Example: Initial Tableau

Example 21 Admissible basis: $\mathbf{A}^{(6)}, \mathbf{A}^{(7)}, \mathbf{A}^{(5)}$

Initial matrix & the 1st simplex tableau:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 1 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(5)} + (\mathbf{A}_{(1)} + \mathbf{A}_{(2)})$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 1 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

Example: Pivot

the 1st simplex tableau & the 2nd simplex tableau:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 1 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(1)} - \mathbf{A}_{(2)}, \quad \mathbf{A}_{(4)} - \mathbf{A}_{(2)}, \quad \mathbf{A}_{(5)} - 2\mathbf{A}_{(2)}$$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \\ 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \end{bmatrix}$$

Example: Further Pivoting

the 2nd simplex tableau & the 3rd simplex tableau ($a_0^* = 0$):

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \\ 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(2)} + \mathbf{A}_{(1)}, \quad \mathbf{A}_{(4)} - \mathbf{A}_{(1)}, \quad \mathbf{A}_{(5)} - \mathbf{A}_{(1)}$$

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Example: Optimal Solution

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Dual LP Problem

Dual Linear Programming Problems

Primal problem:

$$\min \quad \mathbf{c}^T \mathbf{x}$$

subject to

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ \mathbf{x} &\geq \mathbf{0} \end{aligned}$$

Dual problem:

$$\max \quad \mathbf{b}^T \mathbf{w}$$

subject to

$$\mathbf{A}^T \mathbf{w} \leq \mathbf{c}$$

Weak and Strong Duality

Weak Duality Theorem

If $\mathbf{x}$ and $\mathbf{w}$ are feasible solutions of the primal and dual problems, then $\mathbf{b}^T \mathbf{w} \leq \mathbf{c}^T \mathbf{x}$

Strong Duality Theorem

If $\bar{\mathbf{x}}$ and $\bar{\mathbf{w}}$ are feasible solutions of the primal and dual problems, and $\mathbf{b}^T \bar{\mathbf{w}} = \mathbf{c}^T \bar{\mathbf{x}}$, then

- $\bar{\mathbf{x}}$ and $\bar{\mathbf{w}}$ solve their respective problems
- The optimal values of the primal and the dual objective functions are equal:

$$\begin{aligned}\mathbf{c}^T \bar{\mathbf{x}} &= \min \mathbf{c}^T \mathbf{x} \\ &= \max \mathbf{b}^T \mathbf{w} \\ &= \mathbf{b}^T \bar{\mathbf{w}}\end{aligned}$$



Fundamental Duality Theorem

Fundamental Duality Theorem

- If either the primal/dual problem has a finite optimal solution, then the dual/primal problem has a finite optimal solution as well

Moreover,

$$\min \{ \mathbf{c}^T \mathbf{x} : \mathbf{A} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \} = \max \{ \mathbf{b}^T \mathbf{w} : \mathbf{A}^T \mathbf{w} \leq \mathbf{c} \}$$



Proof of Duality ($\mathbf{x} \rightarrow \mathbf{w}$)

Proof ($\mathbf{x} \rightarrow \mathbf{w}$):

- Assume the BFS $x_B = \mathbf{B}^{-1}\mathbf{b}$ and $x_D = \mathbf{0}^T$, the final simplex tableau is

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{B}^{-1}\mathbf{D} & 0 & \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{0}^T & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T & 1 & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \end{bmatrix}$$

$$\mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T \leq \mathbf{0}^T, z_o = \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b}, [\mathbf{I}_m \mathbf{B}^{-1}\mathbf{D}] = \mathbf{B}^{-1}\mathbf{A}$$

- Define $\mathbf{w} = (\mathbf{c}_B^T\mathbf{B}^{-1})^T \in \mathbf{R}_{n \times 1}$

$$\mathbf{A}^T\mathbf{w} = \mathbf{A}^T(\mathbf{c}_B^T\mathbf{B}^{-1})^T = (\mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{A})^T = (\mathbf{c}_B^T[\mathbf{I}_m \mathbf{B}^{-1}\mathbf{D}])^T = [\mathbf{c}_B^T\mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D}]^T \leq [\mathbf{c}_B^T\mathbf{c}_D^T]^T = \mathbf{c}$$

- Thus, $\mathbf{w}$ is a feasible solution of the dual problem

$$\mathbf{b}^T\mathbf{w} = \mathbf{b}^T(\mathbf{c}_B^T\mathbf{B}^{-1})^T = (\mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b})^T = z_o^T = z_o$$

Proof of Duality ($w \rightarrow x$)

Proof ($w \rightarrow x$):

Denote $w = w_1 - w_2$, and slack variables $w_3 \geq 0$

Reformulation of the dual problem:

$$-\min \quad -b^T w_1 + b^T w_2 + 0^T w_3$$

subject to

$$A^T w_1 - A^T w_2 + I w_3 = c^T, \quad w_1 \geq 0, w_2 \geq 0, w_3 \geq 0$$

$$\Downarrow$$

$$-A^T w_1 + A^T w_2 - I w_3 = -c^T, \quad w_1 \geq 0, w_2 \geq 0, w_3 \geq 0$$

Dual of Dual

Proof ($w \rightarrow x$):

The dual problem of the above:

$$\min \quad [-b \quad b \quad 0]x$$

subject to

$$[-A^T \quad A^T \quad -I]x \leq [-c]$$

$$Ax = b, x \geq 0$$



Example: Primal-Dual

Question:

$$\min \quad x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

subject to

$$16x_1 + 2x_2 + x_5 + 10x_6 = 4$$

$$-x_1 + x_4 + x_6 = 0$$

$$5x_1 + 4x_2 + x_3 = 8$$

$$\forall x_i \geq 0$$

Example: Dual Formulation

- The dual problem:

$$\max \quad 4w_1 + 0w_2 + 8w_3$$

subject to

$$16w_1 - w_2 + 5w_3 \leq 1$$

$$2w_1 + 4w_3 \leq 1$$

$$w_3 \leq 1$$

$$w_2 \leq 1$$

$$w_1 \leq 1$$

$$10w_1 + w_2 \leq 1$$

Example: Solving Dual

- The initial simplex tableau:

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 19 & 5 & 0 & 0 & 0 & 10 & 1 & 12 \end{bmatrix}$$

- The final tableau:

$$\begin{bmatrix} 8 & 1 & 0 & 0 & 12 & 5 & 0 & 2 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -27 & 0 & 1 & 0 & -2 & -20 & 0 & 0 \\ -21 & 0 & 0 & 0 & -5/2 & -15 & 1 & 2 \end{bmatrix}$$

- $w^T = c_B^T B^{-1} = c_B^T B^{-1} D - c_D^T + c_D^T$, if $D = I$
 $w_1 = -5/2 + 1 = -3/2$, $w_2 = 0 + 1 = 1$, $w_3 = 0 + 1 = 1$
- The optimal value of the cost function is 2.